Solutions to Exercise Series 1
Fundamental Electronics

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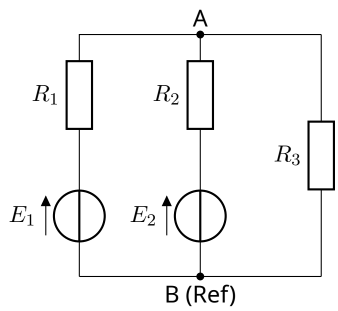
Exercise 1

We aim to determine the current intensities in the three branches of the circuit. Given: $R_1 = 2\,\Omega$, $R_2 = 5\,\Omega$, $R_3 = 10\,\Omega$, $E_1 = 20\,$V, and $E_2 = 70\,$V.



Let the top node be A and the bottom node be B. We can set node B as the reference ground ($V_B = 0\,$V). Applying Millman's Theorem (or Nodal Analysis) at node A to find the voltage V_A:

$$V_A = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{0}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

Substitute the given values:

$$V_A = \frac{\frac{20}{2} + \frac{70}{5} + 0}{\frac{1}{2} + \frac{1}{5} + \frac{1}{10}} = \frac{10 + 14}{0.5 + 0.2 + 0.1} = \frac{24}{0.8} = 30\,\text{V}$$

Now, we can calculate the current in each branch using Ohm's law:

- Current from E_1 (let's call it I_1 flowing towards A):

$$I_1 = \frac{E_1 - V_A}{R_1} = \frac{20 - 30}{2} = -5\,\text{A}$$

(The negative sign indicates the current actually flows from node A towards the source E_1.)

- Current from E_2 (let's call it I_2 flowing towards A):

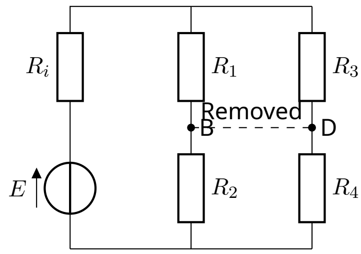
$$I_2 = \frac{E_2 - V_A}{R_2} = \frac{70 - 30}{5} = 8\,\text{A}$$

- Current through R_3 (let's call it I_3 flowing from A to B):

$$I_3 = \frac{V_A - V_B}{R_3} = \frac{30 - 0}{10} = 3\,\text{A}$$

Verification via Kirchhoff's Current Law at node A: $I_1 + I_2 = I_3 \implies -5\,\text{A} + 8\,\text{A} = 3\,\text{A}$ (The currents are consistent).

Exercise 2



Part 1: Network Topology

- **Number of nodes:** 4 (Nodes A, B, C, and D).
- **Number of branches:** 6 (Branches containing R_1, R_2, R_3, R_4, R, and the source branch with E and R_i).
- **Number of independent loops (meshes):** Branches - Nodes + 1 = 6 - 4 + 1 = 3 loops.

Part 2: Currents with BD branch removed

Given $R_1 = R_2 = R_3 = R_4 = 1\,\Omega$, $R_i = 4\,\Omega$, and $E = 10\,\text{V}$. With the central branch BD removed, the circuit simplifies to a source supplying two parallel branches:

- The left branch has R_1 and R_2 in series: $R_{12} = 1 + 1 = 2\,\Omega$.
- The right branch has R_3 and R_4 in series: $R_{34} = 1 + 1 = 2\,\Omega$.

The equivalent resistance of these two parallel branches is:

$$R_p = \frac{R_{12} \times R_{34}}{R_{12} + R_{34}} = \frac{2 \times 2}{2 + 2} = 1\,\Omega$$

The total equivalent resistance of the entire circuit is:

$$R_{eq} = R_i + R_p = 4 + 1 = 5\,\Omega$$

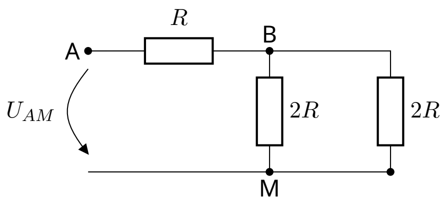
The total current I_{Ri} supplied by the source is:

$$I_{Ri} = \frac{E}{R_{eq}} = \frac{10}{5} = 2\,\text{A}$$

Since the parallel branches have identical resistances ($2\,\Omega$ each), the current splits equally:

$$I_{R1} = I_{R2} = 1\,\text{A} \quad \text{and} \quad I_{R3} = I_{R4} = 1\,\text{A}$$

Exercise 3



1. Determine U_{BM} as a function of U_{AM}

Looking to the right of node B, we have a $2R$ resistor in parallel with another $2R$ resistor. Their equivalent resistance is:

$$R_{eq,BM} = \frac{2R \times 2R}{2R + 2R} = R$$

Using the Voltage Divider Theorem between node A and the reference node M, the voltage U_{BM} across this equivalent resistance is:

$$U_{BM} = U_{AM}\left(\frac{R_{eq,BM}}{R_{AB} + R_{eq,BM}}\right) = U_{AM}\left(\frac{R}{R+R}\right) = \frac{U_{AM}}{2}$$

2. Determine U_{AM} then U_{BM} as a function of E

The total resistance to the right of node A is the series combination of R_{AB} and $R_{eq,BM}$, which is $R + R = 2R$. This entire right-hand section is in parallel with the $2R$ resistor connected directly across nodes A and M. The equivalent resistance at node A is therefore:

$$R_{eq,A} = \frac{2R \times 2R}{2R + 2R} = R$$

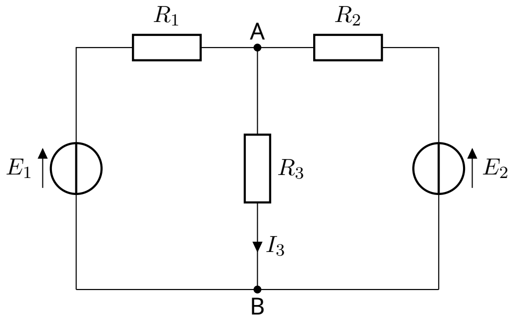
Using the Voltage Divider Theorem from the main source E:

$$U_{AM} = E\left(\frac{R_{eq,A}}{R + R_{eq,A}}\right) = E\left(\frac{R}{R+R}\right) = \frac{E}{2}$$

Substituting U_{AM} into the first relation to find U_{BM}:

$$U_{BM} = \frac{U_{AM}}{2} = \frac{E/2}{2} = \frac{E}{4}$$

Exercise 4



Given: $E_1 = 4\,\text{V}$, $E_2 = 24\,\text{V}$, $R_1 = 16\,\text{k}\Omega$, $R_2 = 4\,\text{k}\Omega$, $R_3 = 6\,\text{k}\Omega$. We need to calculate the current I_3 flowing down through the central AB branch.

Method 1: The Superposition Theorem

Subcircuit 1: E_1 active, E_2 short-circuited. R_2 and R_3 are in parallel: $R_{23} = \frac{4\times6}{4+6} = 2.4\,\text{k}\Omega$. The total current from E_1 is $I_1' = \frac{E_1}{R_1+R_{23}} = \frac{4}{16+2.4} = 0.2174\,\text{mA}$. By the Current Divider Theorem, the current through R_3 (flowing downwards) is:

$$I_3' = I_1'\left(\frac{R_2}{R_2+R_3}\right) = 0.2174 \times \left(\frac{4}{10}\right) = 0.087\,\text{mA}$$

Subcircuit 2: E_2 active, E_1 short-circuited. R_1 and R_3 are in parallel: $R_{13} = \frac{16\times6}{16+6} = 4.364\,\text{k}\Omega$. The total current from E_2 is $I_2'' = \frac{E_2}{R_2+R_{13}} = \frac{24}{4+4.364} = 2.87\,\text{mA}$. By the Current Divider Theorem, the current through R_3 (flowing downwards) is:

$$I_3'' = I_2''\left(\frac{R_1}{R_1+R_3}\right) = 2.87 \times \left(\frac{16}{22}\right) = 2.087\,\text{mA}$$

Total Current: Since both sub-currents flow downwards through the AB branch, they add together:

$$I_3 = I_3' + I_3'' = 0.087\,\text{mA} + 2.087\,\text{mA} = 2.174\,\text{mA}$$

Method 2: Kirchhoff's Laws

Applying Kirchhoff's Voltage Law to the two meshes and Current Law at the top node:

$$\begin{aligned}
\text{Mesh 1: } & E_1 - R_1I_1 - R_3I_3 = 0 \implies 16I_1 + 6I_3 = 4\\
\text{Mesh 2: } & E_2 - R_2I_2 - R_3I_3 = 0 \implies 4I_2 + 6I_3 = 24\\
\text{Node: } & I_1 + I_2 = I_3 \implies I_2 = I_3 - I_1
\end{aligned}$$

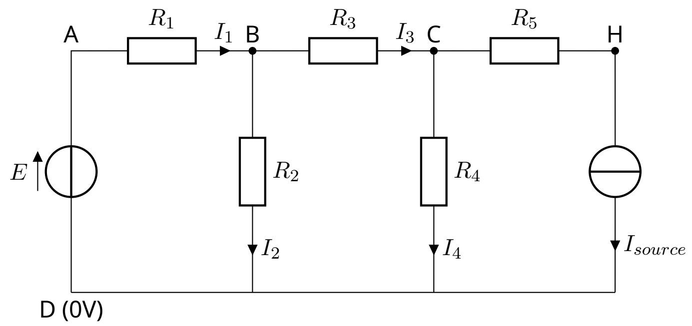
Substitute I_2 into Mesh 2:

$$4(I_3 - I_1) + 6I_3 = 24 \implies -4I_1 + 10I_3 = 24 \implies -16I_1 + 40I_3 = 96$$

Add this derived equation to the Mesh 1 equation to eliminate I_1:

$$(16I_1 + 6I_3) + (-16I_1 + 40I_3) = 4 + 96 \implies 46I_3 = 100 \implies I_3 = \frac{100}{46} = 2.174\,\text{mA}$$

Exercise 5



Given: $E = 10\,\text{V}$, $I_{source} = 0.04\,\text{A}$, $R_1 = 100\,\Omega$, $R_2 = 50\,\Omega$, $R_3 = 100\,\Omega$, $R_4 = 50\,\Omega$, $R_5 = 50\,\Omega$.

1. Superposition: Voltage Source E active, Current Source open

With the current source open, the rightmost branch is disconnected, meaning no current flows through R_5. R_3 and R_4 are now strictly in series: $R_{34} = 100 + 50 = 150\,\Omega$. This is in parallel with R_2: $R_{2,34} = \frac{50\times150}{50+150} = 37.5\,\Omega$. Total current I_1' from E:

$$I_1' = \frac{E}{R_1 + R_{2,34}} = \frac{10}{100 + 37.5} = 0.0727\,\text{A} = 72.7\,\text{mA}$$

Using the Current Divider Theorem:

$$I_2' = I_1' \left(\frac{R_{34}}{R_2 + R_{34}} \right) = 72.7 \times \left(\frac{150}{200} \right) = 54.5\,\text{mA}$$
$$I_3' = I_4' = I_1' \left(\frac{R_2}{R_2 + R_{34}} \right) = 72.7 \times \left(\frac{50}{200} \right) = 18.2\,\text{mA}$$

2. Superposition: Current Source active, Voltage Source E shorted

With E shorted, R_1 and R_2 are in parallel: $R_{12} = \frac{100\times50}{100+50} = 33.33\,\Omega$. This combination is in series with R_3: $R_{12,3} = 33.33 + 100 = 133.33\,\Omega$. The current source draws 0.04 A (or 40 mA) downward from node C. This current splits between R_4 and the $R_{12,3}$ equivalent resistance.

$$I_4'' = -40\,\text{mA} \times \left(\frac{133.33}{50 + 133.33} \right) = -29.1\,\text{mA}$$
$$I_{3(C\to B)} = 40\,\text{mA} \times \left(\frac{50}{50 + 133.33} \right) = 10.9\,\text{mA}$$

Because I_3 is defined flowing from B to C, $I_3'' = +10.9$ mA. This 10.9 mA comes from ground up through R_1 and R_2:

$$I_1'' = 10.9 \times \left(\frac{50}{150} \right) = 3.6\,\text{mA}$$
$$I_2'' = 10.9 \times \left(\frac{100}{150} \right) = 7.3\,\text{mA}$$

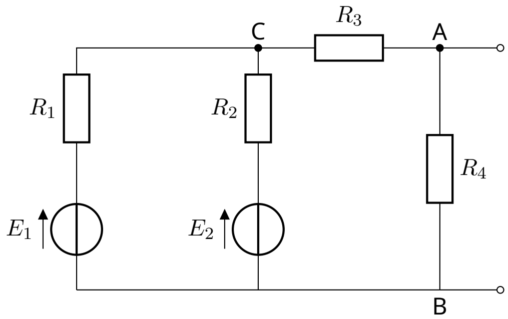
3. Total Currents and Potentials

- $I_1 = I_1' + I_1'' = 72.7 + 3.6 = 76.3$ mA
- $I_2 = I_2' + I_2'' = 54.5 + 7.3 = 61.8$ mA
- $I_3 = I_3' + I_3'' = 18.2 + 10.9 = 29.1$ mA
- $I_4 = I_4' + I_4'' = 18.2 - 29.1 = -10.9$ mA

Potentials relative to node D (Ground, $V_D = 0$ V):

- $V_A = 10$ V
- $V_B = 10 - R_1 I_1 = 10 - (100 \times 0.0763) = 2.37$ V
- $V_C = V_B - R_3 I_3 = 2.37 - (100 \times 0.0291) = -0.54$ V *(Check: $V_C = R_4 I_4 = 50 \times (-0.0109) \approx -0.545$ V)*
- $V_H = V_C - I_{source} R_5 = -0.545 - (0.04 \times 50) = -2.545$ V

Exercise 6



Calculate current I (through an arbitrary load R_c between A and B) using **Thévenin's Theorem**. Given: $E_1 = 10\,\text{V}$, $E_2 = 5\,\text{V}$, $R_1 = R_3 = R_4 = 100\,\Omega$, $R_2 = 50\,\Omega$.

1. Calculate Thévenin Voltage (V_{th})

Remove the load R_c. The voltage V_{th} is the open-circuit voltage at node A (V_A). Let the junction of R_1, R_2, R_3 be node C. Using Nodal Analysis at node A:

$$\frac{V_A - V_C}{100} + \frac{V_A}{100} = 0 \implies 2V_A = V_C \implies V_C = 2V_{th}$$

Using Nodal Analysis at node C:

$$\frac{V_C - E_1}{R_1} + \frac{V_C - E_2}{R_2} + \frac{V_C - V_A}{R_3} = 0$$
$$\frac{2V_{th} - 10}{100} + \frac{2V_{th} - 5}{50} + \frac{2V_{th} - V_{th}}{100} = 0$$

Multiplying the entire equation by 100 gives:

$$(2V_{th} - 10) + 2(2V_{th} - 5) + V_{th} = 0 \implies 7V_{th} - 20 = 0 \implies V_{th} = \frac{20}{7} \approx 2.857\,\text{V}$$

2. Calculate Thévenin Resistance (R_{th})

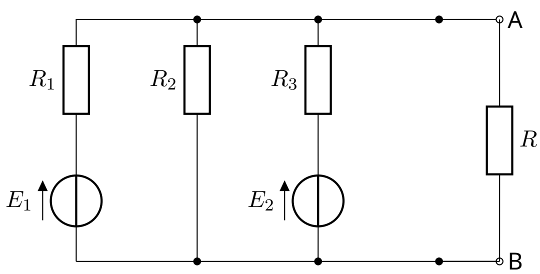
Short-circuit the voltage sources E_1 and E_2. Looking back from terminals A and B: R_1 and R_2 are in parallel: $R_{12} = \frac{100\times50}{100+50} = 33.33\,\Omega$. This is in series with R_3: $R_{12,3} = 33.33 + 100 = 133.33\,\Omega$. This combination is in parallel with R_4:

$$R_{th} = \frac{100 \times 133.33}{100 + 133.33} = \frac{400}{7} \approx 57.14\,\Omega$$

The current through the load R_c is:

$$I = \frac{V_{th}}{R_{th} + R_c} = \frac{20/7}{400/7 + R_c} = \frac{20}{400 + 7R_c}$$

Exercise 7



Given: $E_1 = 10\,$V, $E_2 = 8\,$V, $R_1 = R_2 = R_3 = 200\,\Omega$, $R = 100\,\Omega$.

1. Norton's Theorem

Short the terminals A and B to find the Norton current (I_N). This shorts out the load R and connects all three branch extremities to the reference ground.

$$I_1 = \frac{E_1}{R_1} = \frac{10}{200} = 0.05\,\text{A}$$
$$I_2 = \frac{E_2}{R_3} = \frac{8}{200} = 0.04\,\text{A} \quad \text{(using labels from the provided source document)}$$

The Norton current is the sum of these short-circuit currents:

$$I_N = 0.05 + 0.04 = 0.09\,\text{A}$$

The Norton resistance is found by deactivating the sources:

$$R_N = R_1 \parallel R_2 \parallel R_3 = \frac{200}{3} = 66.67\,\Omega$$

Using the current divider theorem for the load R:

$$I_{AB} = I_N \left(\frac{R_N}{R_N + R}\right) = 0.09 \left(\frac{66.67}{66.67 + 100}\right) = 0.036\,\text{A} \implies V_{AB} = 100 \times 0.036 = 3.6\,\text{V}$$

2. Thévenin's Theorem

The Thévenin resistance is equivalent to the Norton resistance: $R_{th} = 66.67\,\Omega$. The Thévenin voltage can be deduced via source transformation:

$$V_{th} = I_N \times R_N = 0.09 \times 66.67 = 6\,\text{V}$$

The current through the load R is:

$$I_{AB} = \frac{V_{th}}{R_{th} + R} = \frac{6}{66.67 + 100} = 0.036\,\text{A}$$

Both methods consistently provide $I = 36\,$mA and $V_{AB} = 3.6\,$V.